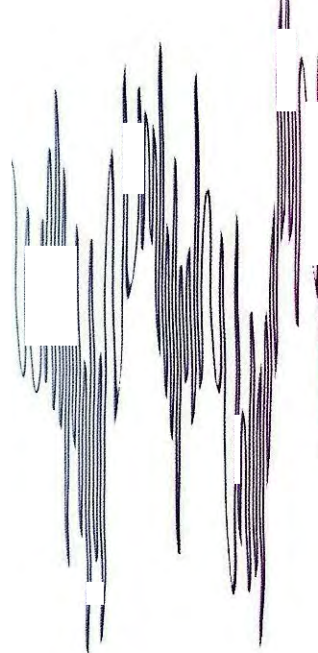






The String Section

The string family is the largest section in the orchestra. It is divided into five subsections made up of four different instruments:

- First Violins: This section is responsible for assuming the highest 'soprano' voice in the section.
- Second Violins: This section also plays the soprano voice but typically in a slightly lower range than the first violins.
- Violas: This section assumes the role of the alto voice.
- Cellos: The cellos act as the tenor voice in the section but can also act as the bass voice.
- Double basses: These are the largest instruments in the string section and they function as the bass voice.

Each of these five string groups or subsections is commonly referred to as a section (i.e. the first violin section, the second violin section, etc.); and is made up of several musicians playing the same part, or in some cases two or more parts. Typically, two musicians sit together at one stand, except in the bass section. The number of players in each section is the number necessary to produce an adequate balance with the rest of the orchestra at all volume levels. The typical string section in a full symphony orchestra consists of the following musicians:

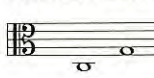
First Violins	16 to 18 players
Second Violins	14 to 16 players
Violas	10 to 12 players
Cellos	10 to 12 players
Double Bases	8 to 10 players

The instruments in the string family all have four strings. The violin, viola and cello are tuned in 5ths and the double bass is tuned in 4ths.

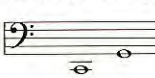
Violin Open Strings



Viola Open Strings



Cello Open Strings



Bass Open Strings

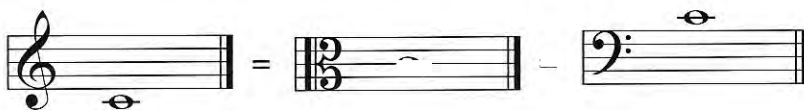


In addition, the range of the double bass can be extended down to a low C by means of an extension device applied to the neck of the bass. This results in the following tuning:

Bass with C-Extension Open Strings



The violin, viola and cello all produce sound exactly as their parts are written. The double bass sounds one octave below the written score. For notation, the violin uses the treble or G clef, the viola uses the alto clef, the cello uses the bass, tenor and treble clefs and the bass uses the bass clef.



In a typical composition, the string section plays more than any other section of the orchestra. They can produce beautiful melodies, extremely lush textures and many different types of accompaniments, and can also be used for a number of unusual effects. Though the strings are extremely homogenous in terms of timbre, they possess a tremendous dynamic range covering *ppp* to *fff*, which allows them to produce a wide variety of performance styles and articulations.

Violin



The **violin** is the most frequently used instrument in the orchestra. Music for the violin is notated on the treble or G clef. Its four strings are tuned to G³, D⁴, A⁴ and E⁵, while the practical range of the instrument is from G³ to E⁷. Each of the strings has a slightly different sound. The instrument has a warm timbre in its lowest range and a vibrant intensity that increases as its upper range is approached. The G string is the thickest and is therefore the most resonant. Its intensity increases dramatically in the higher range. The

D string is warm throughout its range but has less character than the other strings. The A string is most characteristic for its lyrical qualities in its highest range but also fairly brilliant in the first-position range. The E string is the most brilliant and intense of them all. These characteristics become stronger as the upper range is approached. Typically, the first violin section plays in a higher range than the second violin section. However, both sections can play together to achieve increased warmth, intensity or loudness.

The **viola** is a slightly larger instrument than the violin. Music written for the viola is notated using the alto clef, while the treble clef is used to reduce the number of ledger lines in higher passages. The viola's four strings are tuned to C³, G², D³ and A³ and its range is from C³ to approximately G⁶. The viola has a characteristically darker tone than the violin. Its bow and strings are thicker and heavier; therefore it cannot be used to produce as quiet a sound as the violin. The C string is dark as the middle G and D strings are less interesting in timbre. These are used for much of the viola's playing, which often tends to be in an accompaniment capacity. The A string is brighter than the E string on the violin, yet it is very piercing and extremely beautiful. The instrument is excellent for secondary melodies and for melodies that require a slightly different timbre than that of the violins. Violas often double either the violins or cellos in melodic situations. When doing so, the combination is warmer and mellower than when either of the other sections is used by itself.



Cello

The **cello** is the second largest instrument in the string family. Music for it is typically notated using the bass clef. Its four strings are tuned to C², G², D³ and A³—one octave below the viola. Its range is from C² to approximately D⁵. Unlike the violin and viola, which are held under the chin, the cello is held between the seated player's knees and positioned with the neck over the player's left shoulder. The C string is strong in tone while the G string is somewhat weaker. The D string is the warmest of them all and provides an area on which warm, lyrical lines can be played. The A string is piercing and vibrant. The overall tone of the cello is rich in the lower range and incredibly expressive in the higher range, especially above F³. The cello section typically plays melodic lines but it is also extremely adept at playing melodic countermelodies.



The **string bass** (also known as double bass or contrabass) is the largest instrument in the string family. Music for the bass is notated using the bass clef. However, the instrument sounds one octave lower than written. Its four strings are tuned in fourths, to E¹, A¹, D² and G², but notated E², A², D³ and G³. Therefore, when discussing music for the bass, the *written* note is the one that is referred to. As mentioned earlier, a C extension can lower the instrument's range down to C¹ (written C²). The tone of all of its strings is rather non-descript, lacking the elements that give the other three stringed instruments their characteristic sound. The main role of the bass is to play the lines in the lowest range of the orchestra. Predominantly, these lines incorporate the fundamental tones of the harmonic fabric. In orchestral writing up through the time of Beethoven, the bass and cello typically read the same part (hence the name “double bass,” since the bass was doubling the cello an octave lower). In the early 19th

String Bass



century, composers began indicating passages where the cellos were to play while the basses rested, the basses being used more to reinforce the louder passages. By the latter part of the 19th century, it became the norm to provide separate parts for the bass section. It should also be noted that the bass is not very adept at fast or complicated lines.

THE BOW AND ITS USES

With the exception of pizzicato (see below), all articulations are produced on string instruments using the bow. The bow, which gets its name from its visual similarity to the bow used in archery, is made of high-quality horsetail hair stretched tight across a wood shaft or stick. The end of the bow that is furthest away from the player is called the *tip*, while the end closest to the player is called the *heel* or *frog*.

The articulation of a phrase can change dramatically based on how the bow is used. Therefore, if the composer is in mind a specific bowing that he or she feels will better convey the meaning of the phrase, this information should be included in the score. In phrases where the composer feels that a variety of bowings are possible, bow markings may be left out and the choice left to the discretion of the conductor. In talking with many professional string players, I have discovered that most agree that the composer should not put bow markings in the score, leaving it up to the section leader to decide on the bowings. However, it is common practice for the composer to put slur markings in the score, thus denoting what notes are played with the same bowstroke. Even so, according to these same players, the section leaders will often change the details to allow the phrases to be played more efficiently or comfortably.

For those composers writing for real instruments, it is important to have some more detail in regards to the bowstroke. A bowstroke can be played in an upward or downward direction. In this regard, two basic observations can be made. Down-bows tend to sound heavier, since they most often start near the frog area of the bow. This is the area where the hand holds the bow and therefore the player can use more pressure to produce a slightly heavier tone. In addition to the hand placement, the natural movement of pulling the bow downwards contributes to the heavier sound. Up-bows, on the other hand, tend to sound 'lighter' because of the natural upward motion as well as the fact that an up-bow is generally played with the bow positioned further from the frog, where less hand pressure can be applied.

Now mind you, these are generalizations. A good player can make either bowing sound virtually identical. Consequently, choosing whether to use a down- or up-bow often comes down to what will 'feel' best to the musician. For instance, an accented

note is much easier for the musician to play with a down-bow than an up-bow. Similarly, an anacrusis or upbeat is easier to play with an up-bow. Determining where to use up-bows and where to use down-bows is a bit like solving a puzzle. If you have a particular note or passage that requires a specific bowing, then you must work backwards from there to figure out the bowings that come before, so that as a note or passage comes about, it will be in the correct bowing, be it up- or down-bow. Here are some basic guidelines for determining what bowing to use in a given situation.

- Many Western compositions (meaning music of Europe and the Americas) accentuate the first and third beats. Consequently, a down-bow is most often used at the beginning of a measure. The exception to this is when you are trying to create a fluid, non-interrupted line. In that case, there is less of an accent on the first and third beats (or no accent at all), which then allows you to use either bowstroke with equal effectiveness.
- An upbeat or an anacrusis typically begins with an up-bow.
- When you want a crescendo, it makes more sense to accomplish this with an up-bow, since, as the bow travels from tip to frog, the player will be applying more and more pressure and thereby making the sound louder. Conversely, when you want a decrescendo, a down-bow is typically used.
- If you want heavy accents over consecutive notes, repeated down-bows work the best. However, such notes are always slightly separated, because the player needs a moment in which to “retake” the bow, bringing it back to the frog in preparation for the next stroke.
- The quieter the music, the less amount of bow is required to produce a tone. This allows for more notes to be played with one bowstroke. Conversely, the louder the music, the more bow is used, which results in fewer notes that can be played with one bowstroke. Also, notes played in slower tempi use longer bowstrokes, whereas notes played in faster tempi use shorter bowstrokes.

MANNERS OF BOWINGS AND PERFORMANCE TECHNIQUES

There are a number of different manners of bowings and playing or performance techniques. Because of the complexities involved in describing these terms, I felt it best to enlist the help of various professional string players and teachers from across the U.S. Upon discussing the terminology with these

individuals, I came to the realization that the definitions for each of these people were dependent upon where they studied and who they studied with and who they have performed under, especially in an orchestral situation. It is of further interest to me that several of these terms, which inundate every orchestration book I have ever encountered, are not frequently used by string players at all. The following information is my attempt to convey meaning to the terms associated with string bowings and articulations.

SLURRED AND SINGLE BOWED NOTES

When a group of notes is played with single bowstroke, they are referred to as **slurred notes** or a *slurred phrase*. Many string players also refer to this as “legato,” since this is typically the feel that is desired when the notes are connected. For written notation, the composer would place these notes on the staff and group them together under a slur marking. All notes under the slur are played together with one bowstroke in a legato (connected) manner. When groups of notes under separate but consecutive slur markings are encountered, the player bows the first set with one bowstroke and bows the next set with another bowstroke in the opposite direction.



When separate bowstroke is used for each note, regardless of the dynamic or tempo, the notes are said to be **single bowed**. Any articulation that is not slurred is considered single bowed. Single bowed notes can be long, short, accented, non-accented, etc.

The example below illustrates slurred and single bowed notes. The first four notes are joined together with a slur marking and are therefore played with one bowstroke. The second four notes have no slur marking and are therefore played using separate bowstrokes.

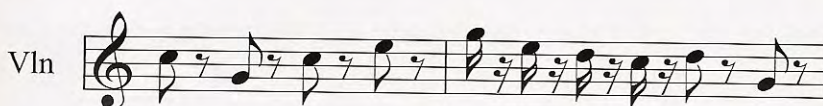
ARTICULATIONS FOR SINGLE BOWED NOTES

If single bowed notes are played between slurred notes, as in a melodic situation, then these are played in one style, typically joined together smoothly without accent. In this instance, this playing style can be referred to as the “notes being taken **détaché**,” though you will tell you that this is more of an educational word than one that is used by most string players. These notes taken *détaché* are typically played in the middle of the bow so that the tone produced is mellow and even, but if the tip of the bow (*a la pointe*) is used, a lighter and more delicate sound is heard. If a heavier and richer *détaché* tone is desired, the frog of the bow is used (*au talon*). There is no special notation for *détaché* except the absence of slur markings.

When single bowed notes require separation between them, then the **staccato** articulation is used. The name is derived from the Italian word *staccare*, which means to detach or separate. A staccato note is played by starting and stopping the bow while it is resting on the string. The note length is shortened in accordance to the tempo and character of the phrase. This results in a clear separation between the notes. Staccato notes are notated by placing a dot over or under the note head. The staccato bowings used in this example:

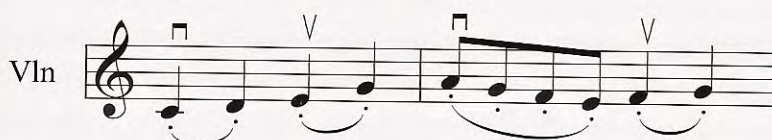


will sound approximately like this:



But all in all, the exact length of the notes is dictated by playing style, tempo and written note length.

A slightly different staccato bowing is the **slurred staccato**. This is performed by slightly separating a series of notes that are all played with one bowstroke. This results in a slight separation between each note. A slurred staccato is notated as staccato notes with a slur mark over them. In this example the first two notes are taken with a downbow, the next two with an upbow, the next four with a downbow and the final two with an upbow.



The phrase will sound like this:



SPECIFIC BOWING TECHNIQUES

the *détaché* and *staccato* terms refer to styles, there are several techniques that are used to produce a variety of articulations. **Martelé** is a heavy, well-accented bowstroke that is normally used in dramatic phrases in louder dynamics. When playing this articulation, the bow moves quickly and stops abruptly while it is left on the string in between notes. Each new stroke is begun with a heavy accent. The result is a sound that has a clean separation between the notes. *Martelé* is used in *marcato* phrases and it is notated using dots, arrowheads, accents or a combination thereof.

is separating the notes as the continues in the same direction. There is a slight the on each note that a attack and release. It is notated as slurred notes dashes or under the note heads. sound is very expressive and is often used in repeated notes for accompaniments, especially in slower tempi.



is a technique in which the bow bounces from note to note. In its simplest form, it can be seen as an off-the-string *staccato* bowing. *Spiccato* is accomplished by decreasing the hold and pressure the right hand on the and the wrist to drop the to the string motion. The bow is then bounced off the string to prepare for the next stroke, which is in the opposite direction. sound from this method is light and delicate in nature.

is a variation of *spiccato* in that there is no "conscious effort" made by the player to raise the the string. the on its the is to as

is an articulation that is without the the strings the instrument with the fleshy part the index finger. notate this within a composition, the word *pizzicato* or *pizz.* must be written into the parts. When the bow is to be used again, *arco* is written into the score. must to a the finger the to rest it into between the thumb and Both these moves take as much as to the be to a rest the performance styles to accommodate this.

variation the *pizzicato* is the was originally introduced in the music Hungarian composer Béla Bartók. The Bartók *pizzicato* is accomplished by pulling the string up away from the

fingerboard and releasing it so that it snaps against the fingerboard. This produces an aggressive popping sound. The *Bartók pizz* has been used in film music and contemporary classical music for decades. Its characteristic sound is useful in certain circumstances, but its overuse can quickly render it ineffective.

There are two types of **tremolos**. The first is a bowing technique and is what most people think of when the term tremolo is used. It is accomplished by keeping the bow on the string while using short and fast up- and down-bowstrokes on a single pitch. This produces a very intense sound that is frequently used in *ff* or *fff* passages. It is a great bowing to use when you need a sustained tone but in a very loud dynamic. The tremolo is also very effective in quieter passages, where it gives a somewhat haunting or mysterious quality to the sound.



The second style is a not a bowing technique but a performance technique. The **fingered tremolo** is produced by drawing the bow across the string while rapidly fingering a repeated interval such as a third, fourth or fifth. The examples below show how these are notated. The fingered tremolo is given a specific time value (such as a quarter-note, half-note, etc.) The values of *each* of the two notes that make up the tremolos are equal to the value of the *entire* tremolo. So if the tremolo is to be played for one beat in 4/4 time, then both notes that make up the tremolo will be quarter notes. However, the notes within the tremolo are not measured. The two notes are simply played quickly in the same manner that a trill would be played. This produces a unique sound that is typically used in quieter sections.



AVANTE-GARDE AND UNUSUAL BOWINGS

While most bowings occur on the violin at its “sweet spot” between the bridge and fingerboard, two bowings take advantage of the timbre differences that occur when the bow is positioned in other areas. ***Sul tasto*** bowing produces an unusual flutelike tone and is accomplished by pulling the bow across the strings *over the fingerboard*. ***Sul ponticello*** is produced by bowing at the lowest ends of the strings toward the bridge. This produces a haunting, glassy sound full of upper harmonics.

Col legno tratto means to play with the wood of the bow, but in a bowing fashion, using the wood instead of the horsetail hair. ***Col legno battuto*** means to play the strings with the wood of the bow by striking the string. This is the more common of the two and as such, it is often indicated simply *col legno*.



OTHER PLAYING CONSIDERATIONS

- *In addition to playing one note at a time, strings can play double stops (two adjacent strings at one time) and in certain circumstances they can play triple stops (three adjacent strings at one time). In the full orchestra, double stops are usually divided between the two players at one stand. The players on the outside of the stand (those nearer the audience) take the upper notes and the players sitting on the inside play the lower notes. To indicate that this is what the composer wants, the word *divisi* or *div.* is inserted into the score. However, if the composer wants each player to play both notes (referred to as a double stop), he or she will write “non div.” in the score. If triple stops (three notes) are to be played at a softer dynamic than *fff*, then the player must roll the chords slightly. If the rolled chord is not desired, then the composer should mark the notes as “divisi.” When a rolled triple stop is being played, the lower notes sometimes anticipate the beat slightly.*
- *Because bow changes can be staggered among the players, long sustained notes that are virtually continuous are possible within the string section. Because this is such a common occurrence in commercial music, the composer might notate this intent by simply putting the word “stagger” or “no breaks” in the score. Usually, this is associated with tied whole notes that extend over several measures. By having the players change bowings in this random fashion, a sustained note with no interruption is produced.*
- *A good section can change bow direction with little break. This can render uninterrupted, smooth flowing legato lines.*
- *String players routinely use vibrato, which enhances the tone of the instrument. This is accomplished by first pressing the finger against the string and then quickly rocking the finger back and forth to slightly raise and lower the pitch of the note being played. Vibrato is usually added to long tones, and because all of the musicians within each section apply vibrato in a slightly different way, the overall sound produced is complex, rich and beautiful. For a plainer, more ethereal sound, the arranger can write “non-vib” in the score.*